# Supplement to 'Toward a "Standard Model" of Early Language Learning'

George Kachergis, Virginia A. Marchman, and Michael C. Frank

August 18, 2021

In the standard model framework we would like to keep real, measurable units on all parameters, so it is important to use empirical values where possible in order to ground the fitted parameters in interpretable units. In the following, we report how we estimate 1) children's average rate of experienced input (tokens of child-directed speech), and 2) the expected number of tokens per word heard by children in a typical month. We then provide details of the models reported in the paper, as well as a few simpler baseline models. All of the analysis scripts, data sources, and fitted models are available on OSF.[1]

Finally, we offer a schematic of how the standard model could be extended to model the differential contributions of child-directed speech and adult-directed speech to children's early word learning, were a suitable dataset available.

## Children's Experienced Rate of Input

To estimate the average hourly input rate of child-directed speech experienced by children, we use the distribution of rates measured from 113 individual children from Hart and Risley (1995), Sperry et al. (2018), and Weisleder and Fernald (2013). The rates from the children in Hart & Risley (1995) were copied from the reported tables, and the remaining datasets were shared with us by the authors, for which we are grateful. Figure S1 plots the distribution of hourly rate of child-directed speech tokens from the primary caregiver (left, $M = 1198$ tokens/hour), as well as the higher rate overheard speech from all adults (right; $M = 1798$), which was only available from 83 children (including 12 additional children from Soderstrom and Wittebolle (2013) for whom caregiver child-directed tokens were not available). Although ultimately we would like to estimate the individual contributions of both child-directed speech and overheard speech from all adults to children's word learning in future iterations of the standard model (see below section on Standard Model 2.0), for now we use the average rate of primary caregiver's child-directed speech, $M \approx 1200$ tokens/hour, since child-directed speech is typically a better predictor of vocabulary growth. Assuming that the average child experiences 1,200 tokens per hour of child-directed speech, 12 hours per day, and 30.44 days per month, they will hear on average 438,336 tokens each month. We can use this average estimate of input in two ways: 1) in combination with each child's fitted latent ability to estimate their amount of experienced input, and 2) in combination with corpus-estimated word frequencies to estimate the number of monthly tokens per word that are experienced by the average child, as we explain in the next section.

## Word Frequencies from CHILDES

To estimate how often children hear particular words, we used a selection of child-directed speech from CHILDES (MacWhinney, 2000). We retrieved transcripts of American English utterances using the childes-db R package (Sanchez et al., 2018) and tabulated $7.680345 \times 10^6$ tokens of 45598 word types. The majority of transcripts were from children between two to four years of age, and were collected during hour-long

[1] https://osf.io/c5u49/

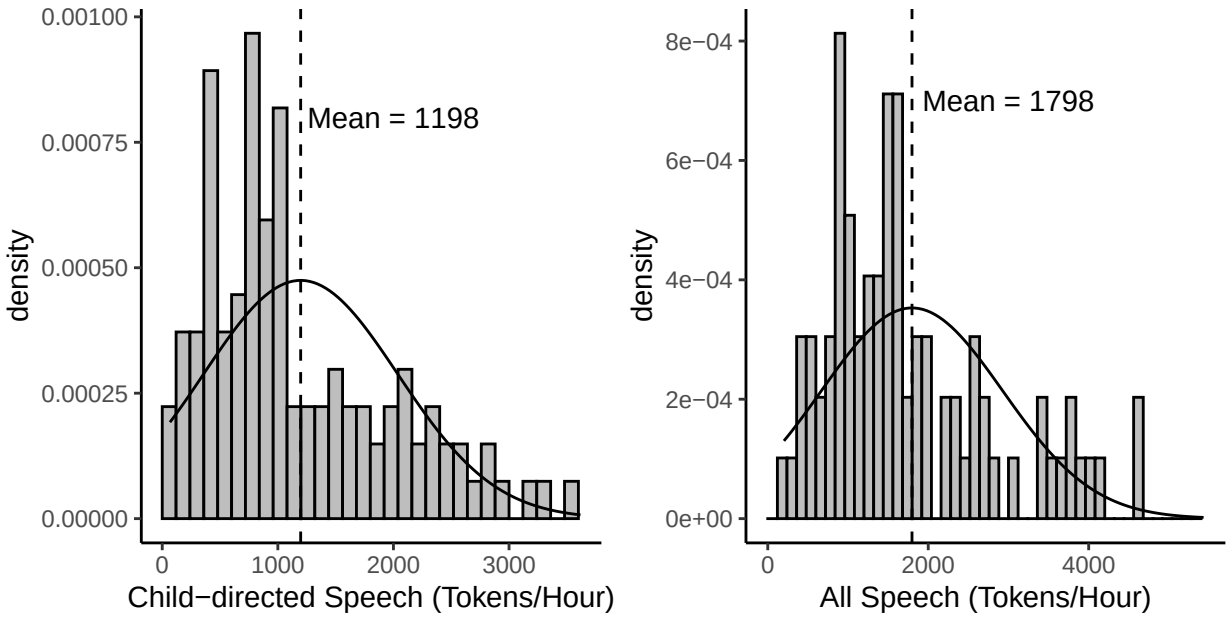


Figure S1: Distribution of hourly child-directed speech rates heard by individual children from their primary caregivers (left) and from all adults (right). On average, two out of three words children hear are child-directed – but individual children's average ratios (and overall rates) likely differ.

at-home recording sessions. We further processed the word counts, combining counts for some variants (e.g., "Mom", "mama", "momma"), removing non-words (e.g., "www") and removing any *hapax legomena* (words appearing only once; e.g. "'s", "A_a"; see file get_childes_freqs.R for details). After this processing, the lexicon contained 27673 unique words that appeared a total of $7.646081 \times 10^6$ times in the transcripts, with the median word appearing 6 times. We mapped 659 of these words to words on the CDI Words & Sentences (CDI:WS) form (see preprocess_word_freq.R for details). The 21 CDI:WS words that did not appear in the CHILDES lexicon were assigned the minimum value in the corpus (count = 2). The word counts were converted to probabilities of occurrence: e.g., the most frequent word "you" occurred 340,268 times in the $7.646081 \times 10^6$-word corpus, and thus for a randomly-selected word $w$ from the corpus, the probability that $w$ is "you" was $P(w) = 0.0445$. In the models we further convert the $P(w)$ for each word to an expected number of tokens per month based on the above estimated average of 1200 hourly child-directed tokens experienced by children (438,336 child-directed tokens per month). So for example, the average child can expect to hear "you" $0.0445 \times 438336 = 19507$ times in child-directed speech in a typical month, but will only hear "ball" an average of 331 times, and "dog" on average 232 times. Figure S2 shows the overall distribution of expected monthly tokens for the 680 CDI items by lexical class, estimated from child-directed speech in CHILDES. The average child can expect to hear the median words ("doll" and "lion") 58 times each month by the average child, although the frequency distribution is of course right-skewed (mean = 419).

It should be noted that these rates are sure to vary greatly both within children (month-to-month), as well as across children (e.g., perhaps especially if they have a dog). The large amount of expected per-word variation is why, if we want to explain per-child variation in word learning, it is critical to measure not only children's overall input rates, but also the rates of each individual word in the input.

Although we have focused thus far on both the number of tokens of and prevalence of words in child-directed speech, future versions of the Standard Model should take into account two additional sources of input variations. First, children also overhear speech that is not directed at them (other-directed speech; ODS): see the distribution of all daily tokens heard by children at right in Figure S1. Second, the frequency with which particular words appear in child-directed vs. other-directed may vary significantly (e.g., we presumably talk less with our children about 'taxes', and more with them about 'toys'.) See below discussion of Standard

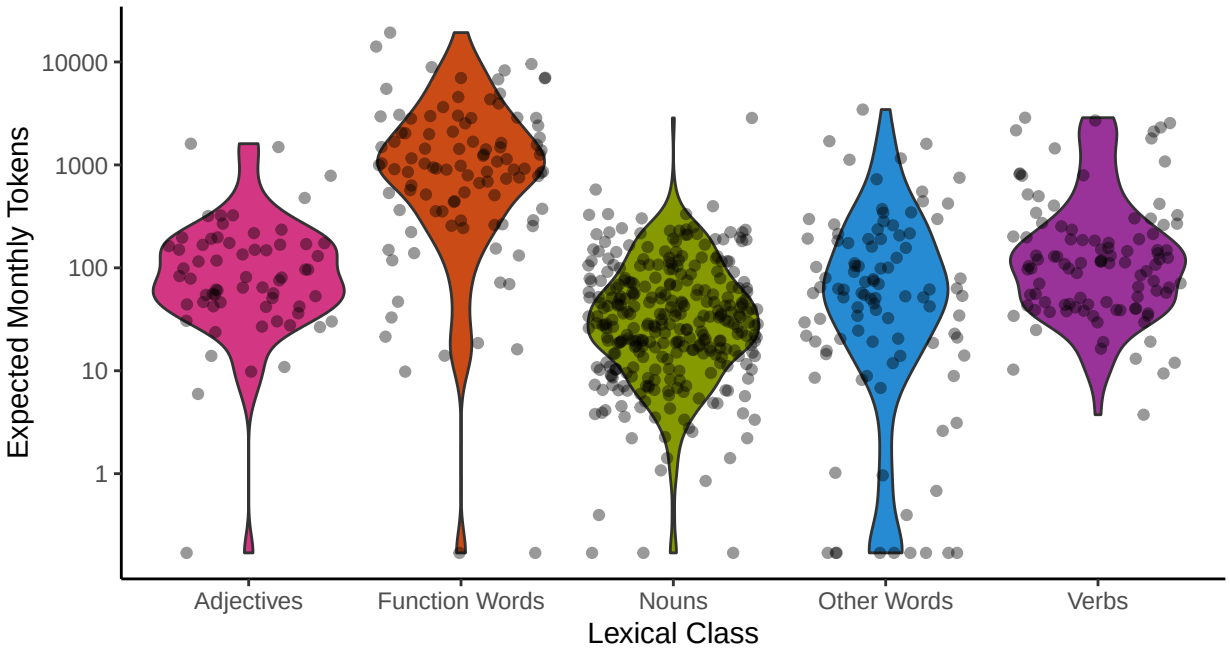


Figure S2: Distribution of expected monthly tokens for the 680 CDI items by lexical class, based on CHILDES word frequencies.

Model v2.0 for how to estimate contributions of other-directed speech to children's early word learning.

## Models

Finally, we fitted models to American English Wordbank production data (Words & Sentences form) from 5,492 children 16-30 months of age. The following models were fitted in R using the lme4 package:

- **m_1pl**: produces ~ -1 + (1 | item) + (1 | person)
- **m_tokmo**: produces ~ -1 + tok_per_mo + (1 | item) + (1 | person)
- **m_lc**: produces ~ -1 + lexical_class + (1 | item) + (1 | person)
- **m2**: produces ~ -1 + lexical_class * lifetime_tokens + (1 | item) + (1 | person)
- **m3**: produces ~ -1 + lexical_class * tok_per_mo + age + (1 | item) + (1 | person)
- **m4**: produces ~ -1 + lexical_class * tok_per_mo * age + (1 | item) + (1 | person)

The first three models are baseline models: m_1pl is the 1-parameter logistic (Rasch) model, estimating a single latent ability parameter per child and a difficulty per item, both as random intercepts. m_tokmo includes the item-level covariate tok_per_mo which represents the CHILDES-estimated monthly child-directed tokens of each word received by the average child (i.e., a child hearing ~438,000 tokens/month). m2 includes lexical class and the expected number of lifetime tokens (tok_per_mo * age) per word received by the average child, and their interaction. m3 includes lexical class, tokens per month, and their interaction, and estimates a separate age effect. Finally, m4 extends m3 to include possible interactions of tok_per_mo and age, and of lexical_class, tok_per_mo, and age.

Table S1: Model comparisons.

| | npar | AIC | BIC | logLik | deviance | Chisq | Df | Pr(>Chisq) |
|---|---|---|---|---|---|---|---|---|
| m_1pl | 2.00 | 2,253,598.74 | 2,253,625.00 | -1,126,797.37 | 2,253,594.74 | NA | NA | NA |
| m_tokmo | 3.00 | 2,253,584.33 | 2,253,623.73 | -1,126,789.17 | 2,253,578.33 | 16.41 | 1.00 | 0.00 |
| m_lc | 7.00 | 2,253,329.63 | 2,253,421.56 | -1,126,657.81 | 2,253,315.63 | 262.70 | 4.00 | 0.00 |
| m2 | 12.00 | 2,252,621.52 | 2,252,779.12 | -1,126,298.76 | 2,252,597.52 | 718.11 | 5.00 | 0.00 |
| m3 | 13.00 | 2,248,868.06 | 2,249,038.79 | -1,124,421.03 | 2,248,842.06 | 3,755.47 | 1.00 | 0.00 |
| m4 | 22.00 | 2,244,676.63 | 2,244,965.56 | -1,122,316.32 | 2,244,632.63 | 4,209.42 | 9.00 | 0.00 |

Table S2: Coefficients from m3.

| | Estimate | Std. Error | z value | Pr(>\|z\|) |
|---|---|---|---|---|
| lexical_classadjectives | -12.18 | 0.27 | -44.48 | 0.00 |
| lexical_classfunction_words | -13.72 | 0.24 | -58.01 | 0.00 |
| lexical_classnouns | -11.26 | 0.17 | -67.90 | 0.00 |
| lexical_classother | -11.60 | 0.22 | -52.07 | 0.00 |
| lexical_classverbs | -12.01 | 0.23 | -53.12 | 0.00 |
| tok_per_mo | 0.00 | 0.00 | 1.01 | 0.31 |
| age | 0.47 | 0.01 | 81.67 | 0.00 |
| lexical_classfunction_words:tok_per_mo | 0.00 | 0.00 | -0.73 | 0.47 |
| lexical_classnouns:tok_per_mo | 0.00 | 0.00 | 1.21 | 0.23 |
| lexical_classother:tok_per_mo | 0.00 | 0.00 | 1.52 | 0.13 |
| lexical_classverbs:tok_per_mo | 0.00 | 0.00 | -0.49 | 0.63 |

## Model comparison

Model comparisons are shown in Table S1, arranged in order of increasing complexity. The successively more complex models show significant improvement in fit (smaller values of AIC and BIC are preferred). **m_tokmo** offers a small but significant improvement in fit compared to the Rasch model (**m_1pl**). Including lexical class rather than monthly tokens per word further improves fit ( **m_lc**), but including both lexical class, expected lifetime tokens (a person-item covariate), and their interaction in **m2** is even better. **m3**, with lexical class, monthly tokens, their interaction, and a separate effect of age fits better than **m2**. Finally, allowing for interactions of lexical class and monthly tokens with age in **m4** achieves superior fit. However, for the sake of brevity and interpretability we focus on the comparison of m2 and m3 below and in the paper.

## Effects

Table S2 shows the estimated coefficients from m3. All lexical class coefficients are significant, as well as age, but there is no significant effect of monthly tokens per word nor interactions of it with lexical class.

Table S3 shows the estimated coefficients from m4.

m4 revealed significant main effects of lexical class and age, but not of monthly tokens. Significant two-way interactions of age and lexical class and significant three-way interactions of lexical class, monthly tokens, and age show that words' quantities and class influence children's ability to learn them to a varying degree across ages. A non-significant interaction of age and monthly tokens was trending positive, leaving room for future investigations with more sensitive models and larger, perhaps cross-linguistic datasets to seek evidence for changes in overall language ability across development.

Table S3: Coefficients from m4.

| | Estimate | Std. Error | z value | Pr(>\|z\|) |
|---|---|---|---|---|
| lexical_classadjectives | -12.70 | 0.28 | -46.17 | 0.00 |
| lexical_classfunction_words | -14.13 | 0.24 | -59.13 | 0.00 |
| lexical_classnouns | -11.11 | 0.17 | -66.93 | 0.00 |
| lexical_classother | -10.40 | 0.22 | -46.63 | 0.00 |
| lexical_classverbs | -13.25 | 0.23 | -58.33 | 0.00 |
| tok_per_mo | 0.00 | 0.00 | 0.74 | 0.46 |
| age | 0.50 | 0.01 | 81.94 | 0.00 |
| lexical_classfunction_words:tok_per_mo | 0.00 | 0.00 | -0.37 | 0.71 |
| lexical_classnouns:tok_per_mo | 0.00 | 0.00 | 0.69 | 0.49 |
| lexical_classother:tok_per_mo | 0.00 | 0.00 | 0.90 | 0.37 |
| lexical_classverbs:tok_per_mo | 0.00 | 0.00 | -0.29 | 0.77 |
| lexical_classfunction_words:age | -0.01 | 0.00 | -2.19 | 0.03 |
| lexical_classnouns:age | -0.03 | 0.00 | -15.27 | 0.00 |
| lexical_classother:age | -0.07 | 0.00 | -34.45 | 0.00 |
| lexical_classverbs:age | 0.03 | 0.00 | 13.72 | 0.00 |
| tok_per_mo:age | 0.00 | 0.00 | 1.85 | 0.06 |
| lexical_classfunction_words:tok_per_mo:age | 0.00 | 0.00 | -2.38 | 0.02 |
| lexical_classnouns:tok_per_mo:age | 0.00 | 0.00 | 2.79 | 0.01 |
| lexical_classother:tok_per_mo:age | 0.00 | 0.00 | 3.76 | 0.00 |
| lexical_classverbs:tok_per_mo:age | 0.00 | 0.00 | -1.15 | 0.25 |

### Plotting Standard Model Predictions

Below we plot selected predictions from the fitted models, with a focus on m3. Figure S3 shows marginal predicted effects of lexical class and word frequency for a typical ability 23-month-old child. The overall distributions of CDI words' CHILDES frequencies are shown at left in Figure S2.

### Plot Sample of Children and Words

Figure S4 shows predicted acquisition curves from m3 for a small subset of the 680 words (left) and for a random sample of 10 children from the Wordbank sample (right). The learning curves of both individual words and individual children show much variation – for words, based on their estimated difficulty, and for children, based on their estimated language ability. Note that although each of these children were only tested once, the model predicts an entire trajectory per child.

## Standard Model 2.0: Child-directed vs. Overheard Speech

The version of the Standard Model (v1.0) presented above and in the paper uses word frequencies from child-directed speech (from the CHILDES database) and expected hourly rates of child-directed speech (CDS) to estimate the number of monthly tokens heard per word, on average. But this ignores the fact that children also overhear speech between adults, and that although this overheard speech may be less attended or otherwise valuable for learning, it may nonetheless explain a significant portion of variation (see Sperry et al., 2018). Figure S5 sketches how a Standard Model v2.0 could use per-child concurrent measures of child-directed and other-directed speech (ODS) to estimate the relative (per-token) value of each type of speech, and to predict learning of individual words (e.g., from the CDI).[2]

[2]We *would* fit this model, but we do not currently have such a dataset.

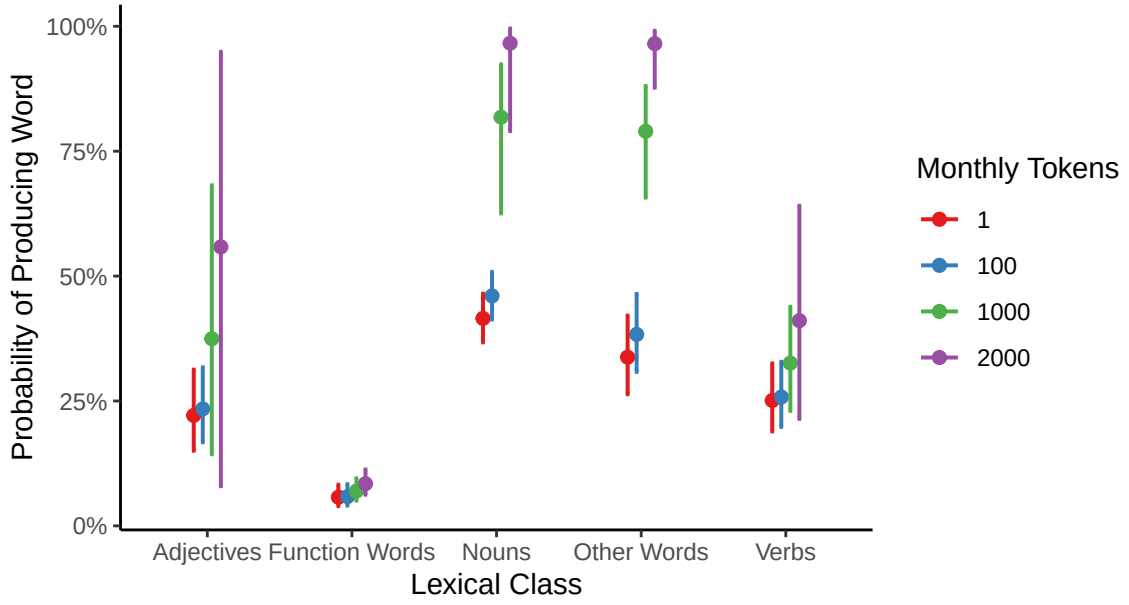


Figure S3: Model m3's predicted probability of the average 23-month-old child producing CDI words of varying frequency and lexical class.

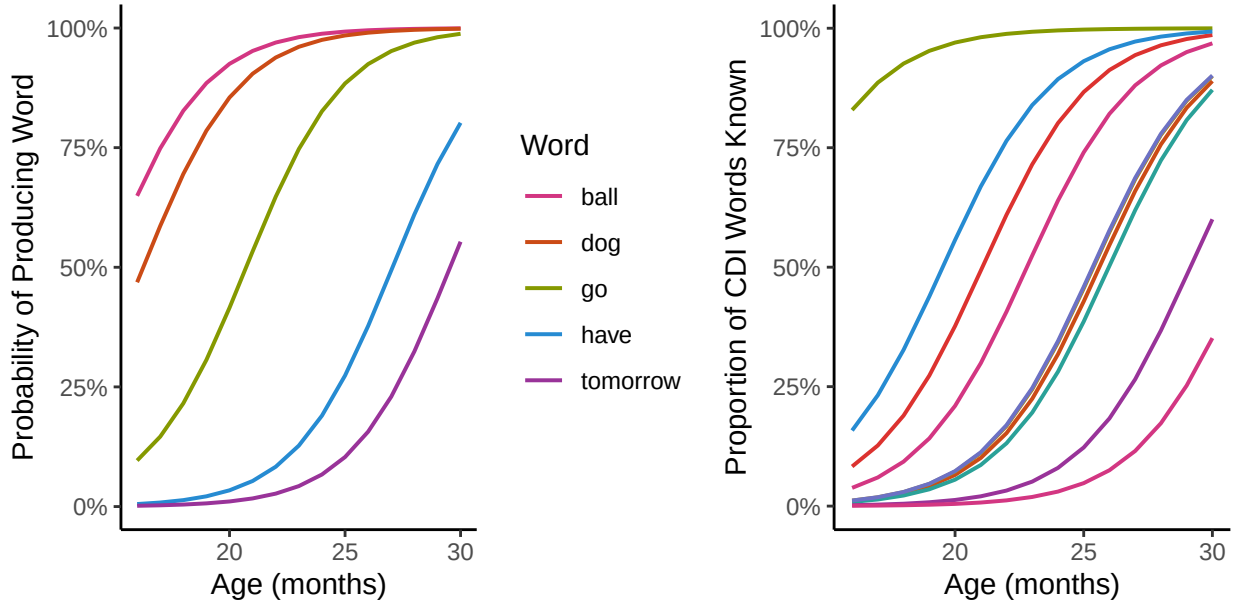


Figure S4: Predicted acquisition curves for a sample of words (left): concrete nouns are learned early while verbs and abstract words (e.g. 'tomorrow') show slower learning. At right are shown predicted learning curves for a sample of 10 children of varying ability.

We show estimated normal distributions for children's hourly rates of input for both CDS ($M$ = 1200 words/hour) and ODS ($M$ = 600 words/hour). Thus, an average child would hourly receive a total of 1800 hourly tokens, with two thirds of it being child-directed speech. In the figure, we show a hypothetical Child 1 who consistently hears 1.5SD more ODS than average (1830 words/hour), and 1SD less CDS than expected (360 words/hour), for a total of (2190 words/hour). An open question is how much more valuable is 1 token of CDS compared to 1 token of ODS, but this multiplier can be estimated if by leveraging input (CDS and ODS) and acquisition data from the same children.

However, it is also important to note that word frequencies sometimes greatly vary between child-directed speech and adult speech, the latter of which we measure using the frequency distribution of 51 million words in subtitles from American English movies (SUBTLEX; Brysbaert & New, 2009). For example, while "you" appears in CDS and ODS at roughly the same rate (44502 tokens/million words in CHILDES; 41857 tokens/million words in SUBTLEX), "ball" is far more frequent in CDS (6631 vs. 105 tokens/million words), and "tomorrow" is more frequent in ODS than CDS (336 vs. 240). Experienced CDS and ODS tokens both contribute to learning each given word (see pie charts at right), but may contribute differentially – not only due to varying word frequency and a child's varying mean rates of CDS and ODS, but due to the relative value of 1 token of CDS vs. 1 token of ODS, which is an estimated parameter of the model.

Extending the Standard Model framework to model bilinguals would schematically look similar (substitute "L1" for CDS, and "L2" for ODS) – although researchers may then want to measure both CDS and ODS in both languages.

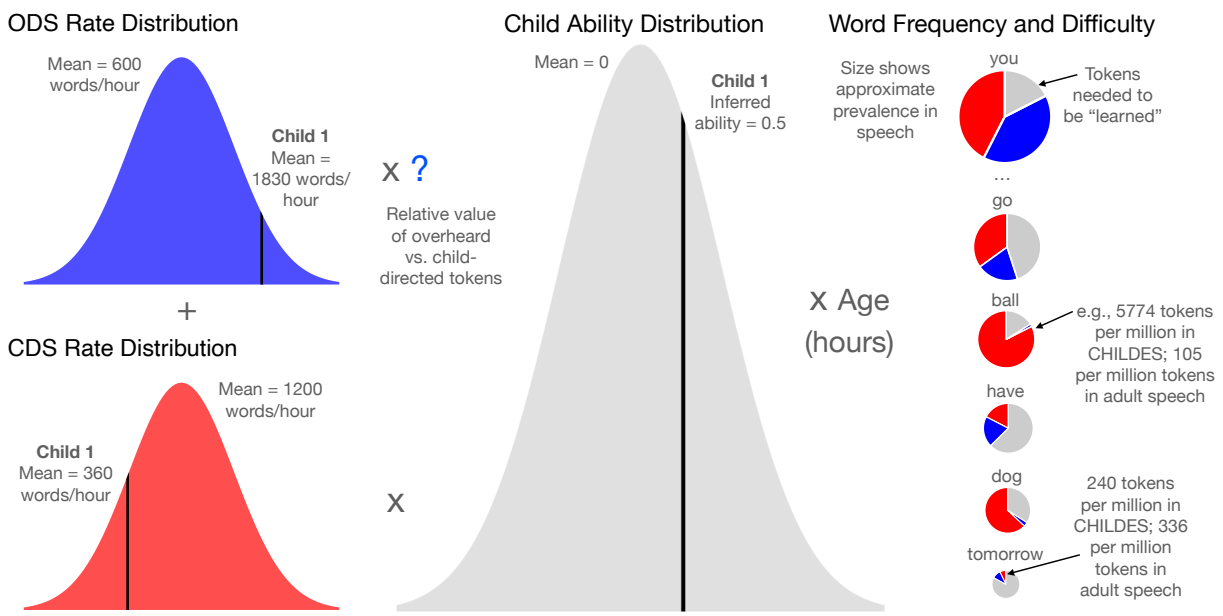


Figure S5: A schematic of how Standard Model v2.0 could incorporate measures of child-directed speech (CDS; red) and other-directed speech (ODS; blue) to also estimate the relative value of the two types of speech. Such a model would also incorporate separate word frequency covariates for CDS and ODS, as word frequencies can vary greatly in CDS vs. ODS (see pie charts at right).